Mohamed Alhaskir

PhD Candidate in Medical AI

I develop and evaluate machine learning methods for clinical data. My current focus is the deployment and monitoring of prediction models in clinical settings; other work includes seizure detection with wearable ECG and LLM-based feedback in medical education.

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EDUCATION

PhD, Medical AI, RWTH Aachen University 2022 – 2027 (expected)

Institute of Medical Informatics · Aachen, Germany

MSc, Biological Information Processing, RWTH Aachen University 2019 – 2022

Major: Computational Neuroscience · Aachen, Germany

BSc, Biological Information Processing, RWTH Aachen University 2016 – 2019

Aachen, Germany

RESEARCH EXPERIENCE

PhD Researcher, Institute of Medical Informatics, RWTH Aachen University 2022 – Present

Aachen, Germany

- MoniPy: Developed a seizure-detection method for wearable ECG combining heart-rate variability features, automatic data quality assessment, and a deep learning classifier. github.com/rwth-imi/monipy
- HaMoJo: Built a local-LLM pipeline that turns audio and video recordings of paediatric simulation training into structured feedback reports, using speech recognition, video-based behavior analysis, and LLM scoring. github.com/Mohamed-Alhaskir/HaMoJo
- FORECAST (AKTIN): Developed machine learning models to forecast hospital resource use, such as inpatient length of stay, from routine emergency department data, and deployed them in the clinical setting with ongoing performance monitoring.

Master's Researcher, Forschungszentrum Jülich (IBI-3) 2021 – 2022

- Master's thesis on spike variation in 3D microelectrode array recordings; developed signal-processing algorithms and optimized the SpyKING CIRCUS spike-sorting pipeline for characterizing spike-waveform variability.

Research Internships, Charité Berlin and RWTH Aachen 2020 – 2021

- Charité Berlin, AG Klinische Neurotechnologie: signal-acquisition and decoding algorithms for an OPM-MEG-based brain–computer interface.
- RWTH Aachen, Molecular & Systemic Neurophysiology: classification algorithms for neural decoding of wide-field calcium-imaging data.

PUBLICATIONS

Under Review

Alhaskir, M., et al. *Trustworthy Artificial Intelligence: Seizure Detection by Wearables Following International Consensus Guidelines.* 2026.

Alhaskir, M., et al. *Automated Feedback Generation in Paediatric Simulation Training: A Prospective Validation of AI-Generated Structured Feedback Reports.* 2026.

Journal Articles and Preprints

Alhaskir, M., et al. *Reliable Detection of Focal Onset Impaired Awareness Seizures in Patients with Epilepsy Using Wearable ECG: Development and Validation Study.* Computer Methods and Programs in Biomedicine, 279, 109295, 2026.

Tayebi Arasteh, S., Farajiamiri, M., Lotfinia, M., Hinrichs-Puladi, B., Bienzeisler, J., **Alhaskir, M.**, et al. *The Role of Self-Supervised Pretraining in Differentially Private Medical Image Analysis.* arXiv:2601.19618, 2026.

Bienzeisler, J., Hertwig, M. K., Heidemeyer, H., **Alhaskir, M.**, et al. *Trans-Sectoral Patient Pathways in Urgent and Emergency Care (TRANSPARENT Study): Protocol for a Prospective, Mixed-Methods Study in Germany.* BMJ Open, 16(2), e114590, 2026.

Conference Papers

Alhaskir, M., et al. *Feasibility of LLM-Assisted Communication Feedback in Simulation-Based Medical Education: A Pilot Study.* Studies in Health Technology and Informatics, 336, 2140–2144, 2026.

Bienzeisler, J., Heidemeyer, H., Röhrig, R., Kombeiz, A., **Alhaskir, M.**, et al. *A Regional Data Space for Urgent and Emergency Care in Germany.* Studies in Health Technology and Informatics, 336, 1167–1171, 2026.

Bienzeisler, J., Hoy, W., Kombeiz, A., **Alhaskir, M.**, et al. *Beating the Clock: A Prediction Model for Timely Care in Emergency Departments.* Studies in Health Technology and Informatics, 316, 1657–1658, 2024.

Alhaskir, M., et al. *ECG Matching: An Approach to Synchronize ECG Datasets for Data Quality Comparisons.* Studies in Health Technology and Informatics, 307, 225–232, 2023.

Alhaskir, M., et al. *Spectral Fusion of Heartbeat and Accelerometer Data for Estimation of Breathing Rate in Wearable Patches.* Studies in Health Technology and Informatics, 302, 1025–1026, 2023.

SKILLS

Programming	Python, SQL, Bash, MATLAB
Machine Learning	PyTorch, TensorFlow/Keras, scikit-learn
LLMs	Hugging Face Transformers, LoRA fine-tuning, vLLM, llama.cpp, evaluation against human raters
Speech & Vision	Whisper, pyannote (speaker diarization), MediaPipe, OpenCV, image classification
Signals & Time Series	ECG and heart-rate variability analysis, sensor fusion (ECG, accelerometer), EEG/MEG, data quality assessment
Data & Statistics	pandas, NumPy, SciPy, statsmodels, SQLAlchemy/MySQL, subject-wise cross-validation
Engineering	Git, pytest, Docker, Slurm (multi-GPU), MLflow, Weights & Biases, FastAPI, Streamlit
Languages	Arabic (native), German (C2), English (C1)

AWARDS & ACTIVITIES

DAAD Prize for Outstanding International Students, RWTH Aachen, 2021

Co-Founder, NeuroTX Aachen e.V., 2020–2022

Developed a brain–computer interface prototype for electric wheelchair control; project funded by the RWTH Aachen Collective Incubator.